

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

03

Q1

ALGEBRA

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$

$$8x + 11y = 170$$

The equation gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x , and the number of 1996 select grade Lincoln Pennies, y , in a collection that is worth a total of \$ 170. If there are 6 1996 select grade Lincoln Pennies in the collection, how many 2009 premium grade Log Cabin Pennies are in the collection?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ALGEBRA

The perimeter of an isosceles triangle is 36 feet. Each of the two congruent sides of the triangle has a length of 10 feet. What is the length, in feet, of the third side?

- A. 10
- B. 12
- C. 16
- D. 18

Q4

ALGEBRA

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

- A. $f(x) = 2x + 3$
- B. $f(x) = 3x + 2$
- C. $f(x) = 4x + 1$
- D. $f(x) = 5x$

Q5

ALGEBRA

Connor has c dollars and Maria has m dollars. Connor has 4 times as many dollars as Maria, and together they have a total of \$ 25.00. Which system of equations represents this situation?

- A. $c = 4m$
 $c + m = 25$
- B. $m = 4c$
 $c + m = 25$
- C. $c = 25m$
 $c + m = 4$
- D. $m = 25c$
 $c + m = 4$

Q6

ALGEBRA

A geologist needs to collect at least 67 samples of lava from a volcano. If the geologist has already collected 63 samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. 130
- B. 63
- C. 4
- D. 0

Q7

ALGEBRA

Line k is defined by $y = 3x + 15$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

A. $-\frac{1}{3}$

B. $-\frac{1}{12}$

C. $-\frac{1}{18}$

D. $-\frac{1}{45}$

Q8

ALGEBRA

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid \$ 2 for the box of cereal and \$ 1.90 per pound for the strawberries. If Lorenzo paid a total of \$ 9.60 for the box of cereal and the strawberries, which of the following equations can be used to find p , the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

A. $1.90p + 2 = 9.60$

B. $1.90p - 2 = 9.60$

C. $1.90 + 2p = 9.60$

D. $1.90 - 2p = 9.60$

Q9

ALGEBRA

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $fx = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of 33 in this context?

- A. The estimated height, in cm, of the liquid at the start of the project
- B. The estimated height, in cm, of the liquid at the end of the project
- C. The estimated change in the height, in cm, of the liquid each day
- D. The estimated number of days for all of the liquid to evaporate

Q10

ALGEBRA

$$y = 4$$

$$x = y + 6$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. 10
- B. 6
- C. 4
- D. 2